

Technical Document #399: Data Engineering - Comprehensive Overview

Technical Document #399: Data Engineering - Comprehensive Overview

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Schema evolution handles changes in data structure over time. Schema registries manage schema versions and ensure backward compatibility.

ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loads it into a target data store.

Data lineage tracks the origin and movement of data through systems. It is essential for debugging, compliance, and impact analysis.

Partitioning divides data into smaller, more manageable pieces. It improves query performance by reducing the amount of data scanned.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Key Takeaways:

- Data warehouses store structured data optimized for analytical queries.
- Data pipelines automate the movement and transformation of data between systems.
- Partitioning divides data into smaller, more manageable pieces.